

Free Open-Source AI Models for English Pronunciation Assessment

Purpose: Help choose a free, downloadable model for building an English pronunciation evaluation system.

At a glance

Model	Main Role	Pronunciation	Offline	Why Use It
Whisper	Speech-to-text	★★	Yes	Best open-source transcription
HuBERT	Speech representation	★★★★★	Yes	Excellent pronunciation features
wav2vec2	Speech representation	★★★★■	Yes	Strong phoneme analysis
MFA	Forced alignment	★★★★★	Yes	Word/phoneme timestamps
SpeechBrain	Toolkit	★★★★	Yes	Easy pretrained models
Kaldi	ASR toolkit	★★★★	Yes	Research/custom pipelines

Recommended Architecture

Audio → Whisper → Transcript → Montreal Forced Aligner → Word & phoneme timestamps → HuBERT (or wav2vec2) → Pronunciation score & feedback

Why Kaldi is no longer the best choice

Designed for researchers: Kaldi is a toolkit, not a plug-and-play model. **Steep learning curve:** Recipes, feature extraction, HMM/DNN pipelines and decoding graphs require significant expertise. **Older architecture:** Modern self-supervised models (HuBERT, wav2vec2) generally provide better speech representations for pronunciation tasks. **More engineering effort:** Building a pronunciation scorer in Kaldi usually requires much more custom work. **Community trend:** Recent academic work has shifted toward HuBERT and wav2vec2 for pronunciation assessment, while Whisper has become the default open-source ASR. **Still valuable:** Kaldi remains an excellent choice if you already have an existing Kaldi pipeline or need maximum control over a traditional ASR system.

When to choose each

Only need transcription? Whisper

Need pronunciation scoring? Whisper + MFA + HuBERT ■

Need phoneme error detection? Whisper + MFA + wav2vec2

Maintaining an old research system? Kaldi + MFA

Official Downloads

Whisper: <https://github.com/openai/whisper>

HuBERT: <https://huggingface.co/models?search=hubert>

wav2vec2: <https://huggingface.co/models?search=wav2vec2>

Montreal Forced Aligner: <https://montreal-forced-aligner.readthedocs.io>

SpeechBrain: <https://github.com/speechbrain/speechbrain>

Kaldi: <https://github.com/kaldi-asr/kaldi>